

Training-Free Few-Shot Personalization for Mobile Mood Inference with TabPFN

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Abstract

Mobile sensing models often generalize poorly to unseen users and geographic populations. DiversityOne showed that a Random Forest Hybrid Model improves mood inference by incorporating labeled target-user examples, but requires fitting a new task-specific classifier for each personalization setting. We investigate whether TabPFN can instead use the same examples through tabular in-context learning without task-specific parameter updates. Using a reconstruction of DiversityOne, we compare TabPFN and the Hybrid Model under matched splits and preprocessing across two valence tasks, six geographic settings, and five personalization budgets (10–50%). TabPFN generally matches or improves on the Hybrid Model, with the most consistent gains in Country-Agnostic I and country-specific three-class prediction, while remaining weaker in some pooled three-class settings. These results establish tabular in-context learning as a strong, setting-dependent training-free baseline for mobile mood personalization.

Resources: DiversityOne dataset [Project Website] | Our code [GitHub]

CCS Concepts

• **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**; *Ubiquitous and mobile computing systems and tools*; • **Computing methodologies** → *Machine learning*; • **Applied computing** → Health informatics.

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Keywords

Mobile Sensing, Mood Inference, Personalization, Tabular In-Context Learning, Geographic Generalization

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1 Introduction

Passive smartphone sensing can support continuous mood inference [9], but models often generalize poorly to users and geographic populations not represented during development [1, 2, 4, 7, 8, 10, 12]. Differences in behavioral patterns and sensing distributions make direct transfer to unseen users and countries challenging.

Prior DiversityOne research showed that incorporating labeled target-user data through a Random Forest Hybrid Model substantially improves mood inference [7]. However, because the Hybrid Model fits a new classifier using source-population and target-user data, it must be retrained whenever the target user, source population, or personalization budget changes. This motivates a training-free alternative for partial personalization.

Recent work has explored large language models (LLMs) for digital mental health, wearable sensing, and affective computing. Mental-LLM shows that pretrained LLMs can support mental health prediction from online text data, but does not address passive smartphone sensing [11]. Health-LLM extends LLM-based prediction to wearable and physiological data, but relies on prompting or fine-tuning [6]. Other recent work suggests that LLMs can also predict affective states from smartphone sensor features [13]. However, these approaches typically convert structured sensing features into textual prompts or adapt language models.

We study a complementary direction: tabular in-context learning for mobile sensing personalization. TabPFN is pretrained for tabular prediction and conditions directly on labeled examples during

inference [5]. Unlike the Random Forest Hybrid Model, TabPFN does not fit a new classifier for each personalization setting. Unlike LLM-based approaches, it operates directly on structured sensing-feature rows rather than textualized prompts. To our knowledge, tabular in-context learning has not previously been studied for personalization in mobile affective and mental sensing. The open question is whether TabPFN can extract more predictive value from limited labeled target-user examples than conventional classifier fitting.

We address two research questions:

- **RQ1:** How does TabPFN compare with the conventional Random Forest Hybrid Model across personalization budgets and geographic modeling settings?
- **RQ2:** Can TabPFN personalize to a new user using only that user’s labeled data, without any source-population data?

For RQ1, we compare TabPFN and the Hybrid Model under matched experimental conditions: identical source-population data, labeled target-user data, fixed held-out target-user test data, feature representations, and preprocessing. We evaluate five personalization budgets: 10%, 20%, 30%, 40%, and 50%. We further evaluate six geographic configurations: country-specific, continent-specific, Country-Agnostic I, Country-Agnostic II, natural multi-country pooling, and balanced multi-country pooling [7]. For RQ2, we compare target-only TabPFN with a target-only Random Forest across the same five support budgets and multiple geographic pooling settings, using identical target-user support and test splits and excluding all source-population data.

This paper makes three contributions:

- To our knowledge, we present the first study of tabular in-context learning for mobile affective and mental sensing.
- We conduct a paired fixed-test evaluation against the established Random Forest Hybrid Model across five personalization budgets, six geographic configurations, and two valence-classification tasks.
- We evaluate target-only personalization across support budgets and geographic pooling settings to determine whether TabPFN can construct personalized predictors without source-population data.

2 Methodology

2.1 Dataset and Prediction Tasks

We use the released DiversityOne smartphone sensing dataset [1], which contains passive sensing and time-diary self-report data collected from participants in eight countries: China, Denmark, India, Italy, Mexico, Mongolia, Paraguay, and the United Kingdom. Our prediction target is mood valence derived from time-diary self-reports.

We evaluate two valence-classification tasks. For the two-class task, valence values 1, 2, 3 are mapped to the positive class and values 4 and 5 are mapped to the negative class. For the three-class task, values 1 and 2 are mapped to positive, value 3 is mapped to neutral, and values 4 and 5 are mapped to negative. The primary metric is macro AUROC. For the three-class task, AUROC is computed using a one-versus-rest formulation. Unless otherwise stated, results are averaged over 10 repeated user-level random holdouts.

In each repeat, approximately 20% of eligible users are randomly selected as unseen target users, and the remaining users form the source population. We report mean macro AUROC with standard error across valid repeats.

2.2 Feature Reconstruction and Filtering

We reconstruct features from the released raw sensing streams. The final feature representation contains approximately 102 features when the required sensors are available [7]. The reconstructed modalities cover location, connectivity, proximity, notifications, activity, screen/touch interaction, user presence, episode duration, and app categories. Following the original DiversityOne feature design, features are aggregated within ± 5 -minute windows around each time-diary report [7].

Our reconstruction differs from the original generalization and personalization study in filtering and preprocessing. The original study reported 678 participants and 329,974 self-reports across eight countries [7]. Our reconstruction first retains valid mood answers, yielding 621 users and 279,585 self-reports; requiring observed sensor data in the ± 5 -minute window leaves 221,197 valid sensing windows. After task-specific label construction and minimum-report filtering, the final evaluated pool contains 526 users and 221,175 self-reports for both two-class task and three-class tasks. Missing values are median-imputed using only training or context rows, without target-test information. We therefore treat our experiments as a public-release reconstruction rather than an exact replication of the original feature table.

2.3 Reconstructed Hybrid Model Baseline

We reconstruct the Random Forest Hybrid Model (HM) from DiversityOne as the reference personalization baseline [7]. HM uses both source-user records and labeled target-support records. When target-support records are added, we remove an equal number of randomly chosen source records (sampled with a per-repeat seed), so that the total number of training or context records stays fixed at the size of that repeat’s source pool. The total is therefore set by the setting and the repeat rather than by the support budget, ranging from approximately 2.6K records in the smallest country-specific configuration to approximately 217.8K records in Country-Agnostic II. This keeps the budget sweep from varying context volume, while the fixed test set isolates the effect of added target-user data. Both models receive exactly the same records under this rule.¹ The Random Forest uses 100 trees.

Our reconstruction reproduces the original personalization effect, although not every absolute AUROC value. Under the filtered pipeline used in our main experiments, aggregate PLM AUROC is 0.531/0.520 for two-/three-class prediction, close to the original 0.510/0.518, while HM reaches 0.699/0.717. When the additional filtering is removed, HM increases to 0.862/0.858, substantially narrowing the discrepancy with the original results. Thus, absolute HM performance is sensitive to reconstruction and filtering choices, while the central personalization effect is consistently reproduced.

¹In 23 of 560 Country-Agnostic I cells, the target-support set exceeds the entire source pool (a small source country paired with a large target country, e.g. India→Italy); there the source pool is exhausted and the context consists of target-support records only. This applies identically to both models, so the paired comparison is unaffected.

2.4 Fixed-Test Target-Support Protocol

We use 10 repeated user-level random holdouts. In each repeat, approximately 20% of eligible users are selected as unseen target users, while the remaining users form the source population. For each target user, a fixed stratified 50% of eligible records is reserved as the target-test set, and the remaining 50% forms the support pool.

We sample target-support records at 10%, 20%, 30%, 40%, and 50% of each user’s eligible records; the 50% budget uses the full support pool. The target-test set remains fixed across budgets. HM and TabPFN use identical source users, target users, support records, test records, features, and preprocessing in each paired comparison.

For statistical comparisons, we form one paired aggregate AUROC difference per repeated user-level holdout and apply a two-sided Wilcoxon signed-rank test across the 10 repeat-level differences.

2.5 Models

Random Forest Hybrid Model. The Hybrid Model uses source-population data and target-support data, and fits a new Random Forest classifier for every repeat, task, geographic setting, and support budget.

TabPFN. We use TabPFN v3 as the tabular in-context learning model in all experiments [3], with two ensemble members ($n_estimators = 2$), a fixed random seed, and the pretraining row limit disabled ($ignore_pretraining_limits = True$), running on a single NVIDIA RTX 4090 GPU. The full source-population and target-support set is passed as the in-context training set in a single forward pass, without subsampling, truncation, or chunking, and the fixed target-test data are predicted as query examples. Realized context sizes range from approximately 2.6K to approximately 217.8K records depending on the geographic setting; the largest contexts exceed TabPFN’s pretraining scale, which we note as a caveat rather than a tuned choice. Features are median-imputed using only context rows, matching the Hybrid Model preprocessing exactly. TabPFN performs no task-specific gradient updates and does not fit a new classifier for each personalization condition [5].²

Target-only Random Forest. For the target-only analysis, the Random Forest is trained from scratch using only the target user’s labeled support data. No source-population data are used.

Target-only TabPFN. The target-only TabPFN model uses the same target-user support data as context and predicts the same held-out target-user test data. It also uses no source-population data and performs no task-specific gradient updates.

2.6 Geographic Settings

We evaluate six geographic configurations, following and extending the DiversityOne evaluation design [7]. Table 1 summarizes how source and target users are defined in each setting.

²We use two ensemble members rather than the default eight for tractability across the full sweep. On a three-country country-specific subset (30 cells, identical splits), raising the ensemble to eight changes macro AUROC by 0.010 on average (max 0.030): two-class results are essentially unchanged (−0.002 mean) and three-class results improve slightly (+0.010 mean). Our setting is therefore conservative with respect to the TabPFN–HM comparison.

3 Results

3.1 RQ1: TabPFN versus the Hybrid Model

We compare TabPFN with the Random Forest Hybrid Model (HM) across five target-support budgets and six geographic settings. All comparisons are paired, using identical source data, target-support data, fixed target-test data, labels, preprocessing, and support budgets. We report $\Delta AUROC$ as TabPFN – HM.

Table 2 shows a setting-dependent pattern. Across repeated user-level holdouts, Country-Agnostic I shows significant TabPFN gains for both tasks at every support budget. Country-specific three-class and continent-specific two-class prediction are also significant across all budgets, while country-specific two-class gains emerge from 20% onward. In contrast, HM significantly outperforms TabPFN for natural multi-country three-class prediction from 20% onward, and Country-Agnostic II remains largely unresolved.

3.2 RQ2: Target-Only Personalization

Table 3 shows that target-only TabPFN generally outperforms a Random Forest trained on the same user-specific support data. Repeat-level tests show significant gains in 33 of 40 setting $\times$ task $\times$ budget cells, including all continent-specific cells and all natural multi-country cells except two-class prediction at 10% support. Country-specific gains are significant in every condition except two-class prediction at 10%, while balanced multi-country results are more mixed.

4 Discussion

4.1 Main Findings

TabPFN is a strong, but not uniformly superior, training-free personalization baseline for DiversityOne mood inference. Its most consistent advantages appear in Country-Agnostic I, country-specific three-class prediction, and target-only personalization. In contrast, HM remains competitive or stronger in several pooled three-class settings, with natural multi-country pooling significantly favoring HM from 20% support onward. Because the paired comparisons use identical source data, target-support data, fixed target-test data, features, and preprocessing, the observed differences primarily reflect the personalization mechanism: HM fits a task-specific Random Forest, whereas TabPFN conditions on the same examples in context [5].

As a temporal robustness check, a chronological country-specific evaluation (earlier 50% for personalization, later 50% for testing) preserved TabPFN’s advantage at 50% support: 0.628 vs. 0.581 (two-class) and 0.657 vs. 0.635 (three-class), with positive differences across all five budgets.

Target-only results show that TabPFN can personalize without source-population data and can sometimes outperform the source-augmented setting, motivating further study of when source context helps or dilutes personalization.

4.2 Limitations

This study is limited to one dataset. Removing the filtering reduced absolute AUROC for both models but changed the TabPFN–RF gap by at most 0.014, providing little evidence that filtering differentially favors TabPFN. At low support budgets, some users lack

Table 1: Geographic configurations used in our evaluation. Source and target users are disjoint under the user-level group 5-fold protocol unless the setting explicitly uses different source and target countries.

Setting	Source population	Target users	Notes
Country-specific	Users from one country	Held-out users from the same country	Per country
Continent-specific	Users pooled within Europe or Asia	Held-out users from the same pooled continent	No Americas
Country-Agnostic I	Users from one source country	Users from a different target country	Ordered pairs
Country-Agnostic II	Users pooled from seven countries	Users from the remaining held-out country	Leave-one-country-out
Multi-country natural	Users pooled from all eight countries, preserving original data volumes	Held-out users from the pooled multi-country dataset	Natural volume
Multi-country balanced	Users pooled from all eight countries after country-level downsampling	Held-out users from the balanced pooled dataset	Balanced volume

Table 2: RQ1 results across target-support budgets. Values are $\Delta\text{AUROC} = \text{TabPFN} - \text{HM}$. Positive values indicate that TabPFN outperforms the Random Forest Hybrid Model. Bold values indicate significant paired differences across the 10 repeated user-level holdouts (two-sided Wilcoxon signed-rank test, $p < 0.05$).

Geographic setting	Two-class ΔAUROC					Three-class ΔAUROC				
	10%	20%	30%	40%	50%	10%	20%	30%	40%	50%
Country-specific	-0.001	+0.029	+0.036	+0.028	+0.030	+0.023	+0.029	+0.030	+0.033	+0.034
Continent-specific	+0.025	+0.025	+0.034	+0.049	+0.042	-0.006	+0.001	-0.010	-0.008	-0.003
Country-Agnostic I	+0.022	+0.022	+0.027	+0.029	+0.031	+0.026	+0.028	+0.028	+0.028	+0.027
Country-Agnostic II	+0.024	+0.018	+0.022	+0.019	+0.011	+0.007	+0.004	+0.006	-0.006	-0.008
Multi-country natural	+0.015	+0.022	+0.025	+0.026	+0.029	-0.010	-0.007	-0.014	-0.019	-0.021
Multi-country balanced	+0.002	+0.006	+0.018	+0.011	+0.023	+0.023	+0.018	+0.017	+0.020	+0.025

Tests use paired aggregate AUROC values from the 10 repeated user-level holdouts. Holm-Bonferroni-adjusted results are provided with the released code.

Table 3: RQ2 target-only personalization results across target-support budgets. Values are $\Delta\text{AUROC} = \text{TabPFN} - \text{RF}$. Positive values indicate that TabPFN outperforms the target-only Random Forest trained on the same user-specific support data. Bold values indicate significant paired differences across the 10 repeated user-level holdouts (two-sided Wilcoxon signed-rank test, $p < 0.05$).

Geographic setting	Two-class ΔAUROC					Three-class ΔAUROC				
	10%	20%	30%	40%	50%	10%	20%	30%	40%	50%
Country-specific	+0.008	+0.033	+0.034	+0.029	+0.036	+0.017	+0.025	+0.029	+0.028	+0.023
Continent-specific	+0.027	+0.030	+0.038	+0.043	+0.040	+0.031	+0.033	+0.030	+0.031	+0.030
Multi-country natural	+0.009	+0.023	+0.030	+0.033	+0.037	+0.020	+0.024	+0.023	+0.023	+0.021
Multi-country balanced	-0.013	+0.005	+0.017	+0.024	+0.026	+0.007	+0.017	+0.024	+0.023	+0.025

Tests use paired aggregate AUROC values from the 10 repeated user-level holdouts. Holm-Bonferroni-adjusted results are provided with the released code.

minority-class examples, particularly for the imbalanced two-class task; we therefore interpret the 10% condition cautiously. TabPFN also required 7.4× the total wall-clock time of HM in our country-specific benchmark; thus, “training-free” refers to the absence of task-specific parameter optimization rather than faster computation. Future work should test additional sensing datasets and tabular foundation models.

Ethical Statement. We use DiversityOne under its approved research-use conditions; mood-inference predictions should not inform high-stakes decisions without further validation and human oversight.

5 Conclusion

Across matched personalization settings, TabPFN provides a strong but setting-dependent alternative to the Random Forest Hybrid Model, with the most consistent gains in Country-Agnostic I, country-specific three-class prediction, and target-only personalization.

These results support tabular in-context learning as a promising training-free personalization approach for mobile affective and mental sensing, without implying uniform superiority over task-specific fitting.

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